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VL-Eraser: Vacuum Distillation for Machine Unlearning in Vision-Language Models

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Abstract

Machine unlearning (MU) aims to remove sensitive or undesired content from pre-trained models. Existing MU methods are commonly characterized as gradually degrading model performance on undesired data to realize approximate forgetting. Despite their successes, the effectiveness in multimodal unlearning tasks remains largely unexplored. In this paper, we first conduct an in-depth analysis and reveal that traditional MU methods tend to disrupt cross-modal alignment, leading to incomplete forgetting in multimodal scenarios. To tackle this challenge, we propose VL-Eraser, a novel unlearning paradigm for VLM unlearning. VL-Eraser reformulates unlearning in VLMs as a two-stage process: distillation and deletion. Specifically, VL-Eraser first introduces a vacuum distillation that disentangles undesired knowledge from the intricate parameters of VLMs and transfers it into low-rank adapters (LoRA). After distillation, unlearning is efficiently achieved by deleting the LoRA parameters from the original model. Extensive experiments across multiple VLM backbones demonstrate that VL-Eraser achieves superior unlearning performance while preserving utility compared to the state-of-the-art baselines.

1. Introduction

Recently, Vision-Language Models (VLMs) have achieved remarkable performance on various applications [13, 17, 36, 42]. As shown in Figure 1(b), by integrating vision encoders [10, 33] with language models in a sequential manner [1, 25, 38, 40], VLMs can leverage multimodal information to tackle complex tasks. However, the rapid development of pre-trained models has raised a series of ethical and security concerns, such as personal privacy [2, 32] and copyrighted data [8, 26]. In response to these concerns, regulations have mandated the Right to be Forgotten, requiring applications to remove private data from models upon request. Machine unlearning has emerged as a feasible solution.

Machine Unlearning (MU) aims to remove sensitive or undesired content from pre-trained models without retraining from scratch [22, 31]. Existing MU approaches typically follow a **reverse-training** paradigm, characterized by inverting the pre-training objective on undesired data to intentionally degrade model performance, thereby achieving approximate forgetting. For instance, methods such as gradient ascent [23, 35], KL-based negative optimization [4, 30], and negative preference optimization [34, 43] all fall under this paradigm and have demonstrated success in unimodal unlearning tasks. Building on these foundations, several studies have attempted to extend such techniques to multimodal unlearning scenarios [7, 21, 39], yet most methods remain dataset-specific and lack generalization.

Despite these advances, it remains unclear whether such unlearning strategies are truly suitable for VLMs. A notable concern arises from the fact that the performance of VLMs depends on two aspects: (1) accurate knowledge in each model, and (2) reliable alignment across modalities. Consequently, when employing reverse-training to degrade VLM's performance for unlearning, a natural question arises: ***Does it truly erase the knowledge, or merely disrupt cross-modal alignment?*** We use visual examples to intuitively highlight the significance of this issue. As shown in Figure 1(a–c), although cross-modal misalignment appears to also prevent VLMs from generating sensitive content related to the input image of "Donald Trump", further analysis reveals that such superficial forgetting inevitably lead to one of two possible outcomes: (1) the overestimated loss prematurely halts training, or (2) even if training continues, the internal language model of VLMs is trained on low-quality data generated by misaligned vision-encoder. Both effects obstruct genuine knowledge removal, posing a risk of knowledge leakage. Therefore, a deep understanding of the nature of forgetting in VLMs is essential for driving further progress in multimodal unlearning.

To this end, we further investigate this issue through experiments (detailed in the appendix). Specifically, we employ two methods to evaluate the forgetting quality (defined as $1 - \frac{\text{unlearned performance}}{\text{original performance}}$ on the forget set) of reverse-training

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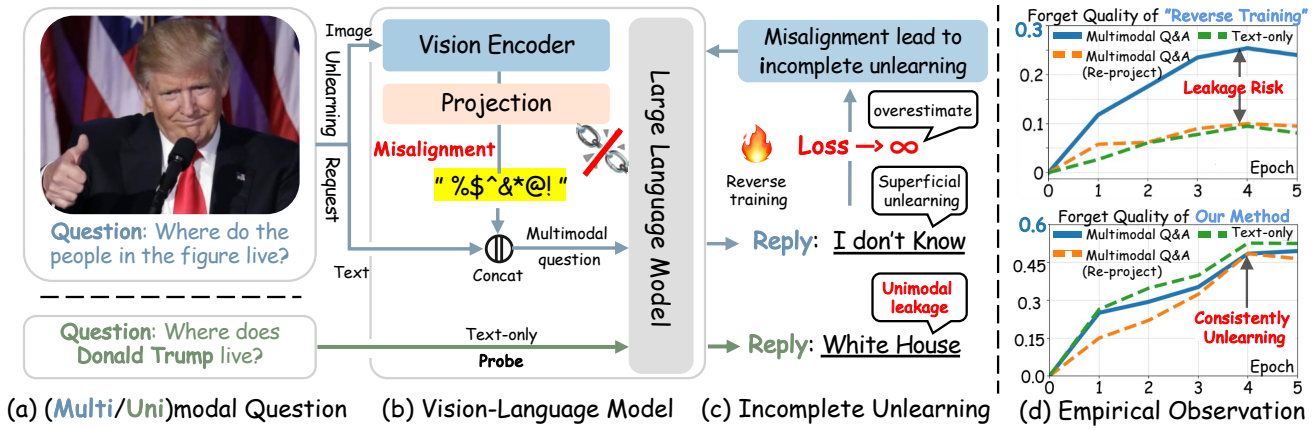


Figure 1. (a-c) Schematic of the traditional unlearning process in VLMs. These methods tend to disrupt modality alignment, hindering genuine knowledge removal. As evidenced in (d), both text-only question probing (- -) and reloading original projection (- -) reveal significant residual knowledge within the LLM, posing a leakage risk. In contrast, our method ensures consistent unlearning in VLMs.

in the VLM: (1) utilizing text-only questions to probe residual knowledge in the language model of the VLM, and (2) reloading the original projection to evaluate residual multimodal knowledge. As shown in Figure 1(d), although the VLM appears to increasingly forget multimodal knowledge over training epochs (blue curve), the internal language model exhibits limited forgetting quality with a flat growth rate (green dash curve), posing a risk of knowledge leakage. Moreover, the reloading-projection method (orange dash curve) reveals that the actual forgetting quality of multimodal knowledge drops to the level of the internal LLM, demonstrating that most of the forgetting stems from disrupted modality alignment rather than actual knowledge removal. In summary, these experiments demonstrate that traditional MU methods tend to disrupt cross-modal alignment, resulting in incomplete forgetting in multimodal settings. This motivates the development of a more reliable unlearning paradigm for VLMs.

To address the above challenge, we propose VL-Eraser, a novel unlearning paradigm for VLMs. VL-Eraser reformulates VLMs' unlearning as a two-stage process: distillation and deletion. Specifically, VL-Eraser introduces a novel distillation method that disentangles undesired knowledge from the intricate parameters of the original model and transfers it into a low-rank adapter (LoRA). To ensure precise knowledge transfer, the distillation is strictly constrained within a vacuum space, leveraging the orthogonality property to naturally filter out irrelevant knowledge. After distillation, the LoRA parameters can be regarded as experts specializing in undesired knowledge. Therefore, unlearning can be efficiently accomplished by subtracting the LoRA parameters from the base model, implying that the corresponding knowledge has been effectively removed. This two-stage design avoids directly degrading the model's performance during training, particularly its cross-modal alignment, leading to better unlearning performance. Extensive experiments demonstrate the superior performance of VL-eraser compared to state-of-the-art models.

Our contributions can be summarized as follows:

- **Issue discovery:** To the best of my knowledge, we are the first to analyze the limitations of traditional machine unlearning paradigms in multimodal scenarios, revealing that they tend to disrupt cross-modal alignment and thus lead to incomplete unlearning.
- **Paradigm innovation:** We propose VL-Eraser, a novel unlearning paradigm for VLMs. VL-Eraser reformulates unlearning as a two-stage process: (1) disentangling undesired knowledge through vacuum distillation, and (2) deleting knowledge via parameter arithmetic. This design effectively prevents the disruption of cross-modal alignment, leading to more comprehensive unlearning.
- **Performance validation:** Extensive experiments across multiple multimodal backbones demonstrate that VL-Eraser achieves superior unlearning performance while preserving utility compared to state-of-the-art baselines.

2. Related Work

In this section, we mainly introduce related work from the following two aspects: Vision Language Models (Section 2.1) and Machine Unlearning (Section 2.2).

2.1. Vision Language Models (VLMs)

VLMs are foundation models that are capable of handling diverse language and vision tasks. Recently, these models have achieved remarkable performance on various multimodal applications [17, 29, 36, 37, 42]. VLMs are typically architected by integrating pre-trained vision encoders [19, 33, 41] with LLMs in a sequential or modular manner. Therefore, the effectiveness of modality alignment directly determines

the performance of VLMs. A pioneering method, Flamingo [1], employs a query-based cross-attention mechanism to establish an advanced and robust vision-language interaction module. In contrast, BLIP-2 [20] employs a lightweight Q-former to more efficiently bridge vision features and LLMs. Additionally, LLaVA [24] demonstrates that simple linear projections can yield strong vision-language reasoning abilities. Besides architectural explorations, several works have highlighted the importance of instruction tuning for strengthening modality alignment. For instance, InstructBLIP [10], LLaVA-1.5 [25], MiniGPT-4 [44], and Otter [18] have developed a system of instructional multimodal data to enhance the models' comprehension capabilities. While VLMs have advanced, multimodal unlearning remains under-explored.

2.2. Machine Unlearning (MU)

Machine Unlearning aims to remove sensitive or undesirable context from the pre-trained models without retraining from scratch [22, 31]. In previous work, the reverse-training paradigm has shown promise for implementing unlearning. Classic methods such as gradient ascent [16, 35] achieve forgetting by degrading performance on undesired data, yet often result in catastrophic loss of overall utility. To address these limitations, subsequent variants have introduced more principled optimization frameworks, such as KL-based negative optimization [4, 30] and negative preference optimization [5, 34, 43]. Unified gradient-based formulations [14] further improve stability and generalization of these unlearning paradigms. In addition, task-vector-based strategies [12, 27] for unlearning have also been explored to eliminate the need for tuning. Compared to single-modality settings, research on unlearning in VLMs remains in its early stages. By applying reverse-training to multimodal fine-tuning pipelines, CLIPErase [39] selectively erases simple visual–textual association in CLIP [6, 33]. SIU [21], Multidelete [7], and MLLMMU [3] further explore unlearning in more complex VLMs. However, these methods are often tailored to particular datasets and are challenging to transfer to the multimodal unlearning benchmark [11, 28]. Furthermore, it remains unclear whether such traditional unlearning paradigm are truly suitable for VLMs.

3. Preliminary

This section begins by introducing the notation and formalizing the problem of VLMs unlearning in Section 3.1. Then, Section 3.2 provides a detailed description of the unlearning objective formulated under the reverse-training paradigm.

3.1. Notion and Problem Setup

Given an initial vision-language model $\mathcal{M}_\theta$, where θ denotes original parameters trained with a dataset $\mathcal{D} = \{v_i, t_i, a_i\}_{i=1}^{N}$ that encompasses N samples, where v_i denotes image, t_i denotes textual instruction and $a_i = (y_1^i, y_2^i, ..., y_{|a_i|}^i)$ denotes the target answer sequence, our goal is to remove the influence of a specific **forget set** $\mathcal{D}^f = \{v_i, t_i, a_i\}_{i=1}^{N_f}$ with N_f visual-textual data samples, while preserving the model's utility on the **retain set** $\mathcal{D}^r = \mathcal{D} \setminus \mathcal{D}_f$. In other words, we would like the model $\mathcal{M}_\theta$ to behave as if the data in the forget set $\mathcal{D}^f$ were never used to train the θ. By definition, the best machine unlearning method is to retrain the model from scratch on $\mathcal{D}^r$ only, which is often intractable in practice due to the prohibitive computational cost.

3.2. Reverse Training Objective

Reverse training is a core concept in many existing unlearning methods, where the key idea is to optimize objectives opposite to those used during the pre-training stage over the forget set $\mathcal{D}^f$.

Therefore, the reverse training objective can be formalized as follows:

$$\mathcal{L}_{unlearn} = - \sum_{(v_i, t_i; a_i) \in \mathcal{D}^f} \mathcal{L}_{pre-train}(\mathcal{M}_\theta(v_i, t_i, a_i)) \quad (1)$$

where $\mathcal{L}_{pre-train}(\cdot)$ denotes the pre-training loss, which is commonly instantiated as the entropy loss (next-token prediction) [35] or KL-divergence constraint term [30]. By inverting the optimization objective of pre-training, the reverse-training-based method intentionally degrades the model's performance on undesired data, thereby achieving approximate forgetting. Despite their successes, the effectiveness of these approaches in multimodal unlearning tasks remains concerning, as they tend to disrupt the cross-modal alignment, leading to incomplete forgetting.

4. Methodology

As discussed above, prior unlearning paradigms typically directly degrade model performance on undesired data, which inevitably disrupts cross-modal alignment, leading to incomplete forgetting. To address this challenge, we propose a novel unlearning paradigm for VLMs called VL-Eraser, which reformulates VLMs' unlearning as a two-stage process: distillation and deletion. In this section, we introduce the key components of our method: Vacuum Distillation (Section 4.1) and Arithmetic Deletion (Section 4.2). The overall framework is shown in Figure 2.

4.1. Vacuum Distillation

Vacuum distillation is the core of our work, which is designed to localize and disentangle undesired knowledge within specific parameters of the original model, thereby facilitating its subsequent knowledge removal. To achieve this, the priority is to divide the original VLM itself into a teacher and student model. Specifically, considering that the feed-forward network (FFN) serves as the primary repository of knowledge in Transformer architectures [9], we only

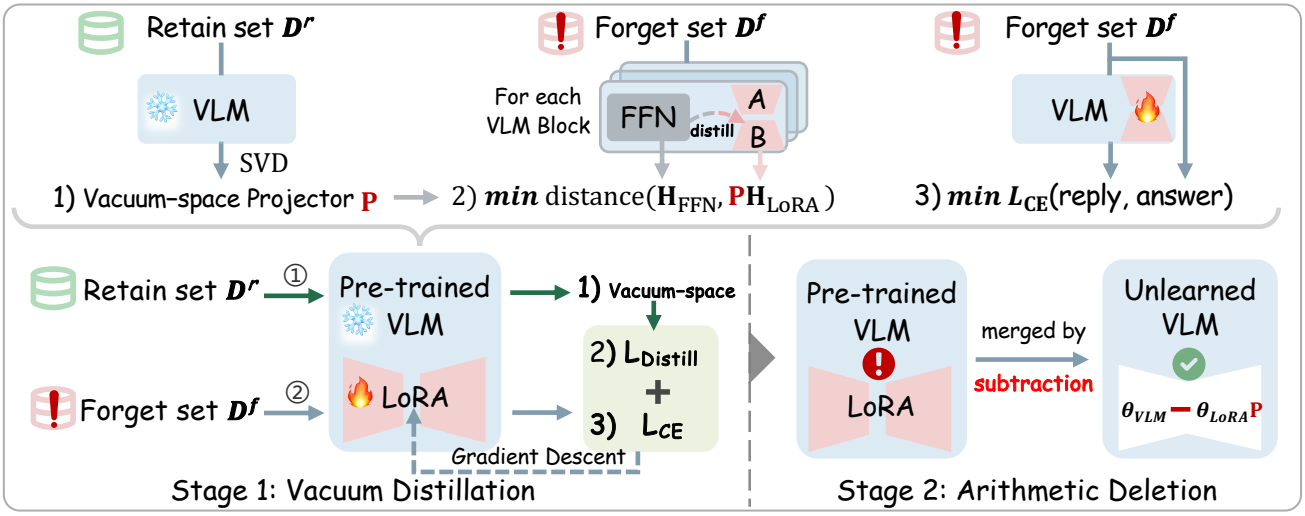


Figure 2. The overall framework of the proposed method VL-Eraser. VL-Eraser reformulates unlearning as a two-stage process. In stage 1, the undesired knowledge is disentangled from the original VLMs and transferred into the LoRA via distillation. To avoid interference from other knowledge, this process is strictly constrained within a vacuum space, which is orthogonal to the retained knowledge. In stage 2, the LoRA parameters encoding undesired knowledge are subtracted from the pretrained model to form a safe yet useful VLM.

apply the low-rank adapters (LoRA) to the FFN of each VLM block. Then, during the distillation training, the frozen FFN parameters serve as the teacher, guiding the student parameters of LoRA to capture the forget-specific knowledge. As shown in Figure 2, the overall vacuum distillation comprises three components described below.

Vacuum Space. Due to the intricate inter-dependencies among diverse knowledge, the precise disentanglement of forget-specific knowledge from VLMs constitutes a non-trivial challenge. To this end, we propose constraining the distillation process within the "vacuum space" of the retain-specific knowledge. Here, we first introduce the definition of the vacuum space (left null space). Specifically, given two matrices X and Y, Y is in the vacuum space of X if and only if $YX = 0$. Motivated by this property, in the context of distillation, projecting the student parameter $\mathbf{W}_{\text{LoRA}}$[1] onto the vacuum space of the retain-specific knowledge representation $\mathbf{H}_r$ yields:

$$Proj_{vacuum}(\mathbf{W}_{\text{LoRA}}) \cdot \mathbf{H}_r = 0, \tag{2}$$

where $Proj_{vacuum}(\cdot)$ denotes the vacuum space projector. $\mathbf{H}_r$ denotes hidden representation corresponding to all input samples from $\mathcal{D}^r$. This helps the projected student parameters naturally remain orthogonal to the retain-specific knowledge, thereby reducing mutual interference during the distillation training.

Next, we describe the implementation of this projection. In image–text multimodal tasks, representation differences are primarily determined by the visual input. To balance efficiency and effectiveness, we thus estimate the vacuum space using intermediate representations of image from the retain set $\mathcal{D}^r$, rather than from all visual–text samples in the $\mathcal{D}^r$. Specifically, we first apply Singular Value Decomposition (SVD) to the image representation $\mathbf{H}_{\text{image}} \in \mathbb{R}^{d \times p \times n}$, where n denote the number of image in $\mathcal{D}^r$, p denotes the number of patch and d denote the hidden dimensions. This process can be formulated as follows:

$$\{U, \Sigma, (U)^\top\} = \text{SVD}(\text{pooling}(\mathbf{H}_{\text{image}})), \tag{3}$$

where pooling$(\cdot)$ denotes the mean-pooling function applied at the patch level, $U \in \mathbb{R}^{d \times d}$ is an orthogonal matrix whose columns are the eigenvectors of the image representation and $\Sigma \in \mathbb{R}^{d \times d}$ is a diagonal matrix containing its singular values, Then, we remove the eigenvectors in U that correspond to non-zero eigenvalues, and define the remaining submatrix as $\hat{U}$. Based on this, the vacuum space project matrix can be defined as $\mathbf{P} = \hat{U}\hat{U}^\top$. Therefore, the vacuum space projector can be formulated as follows:

$$Proj_{vacuum}(\mathbf{W}_{\text{LoRA}}) := \mathbf{W}_{\text{LoRA}}\mathbf{P}. \tag{4}$$

This projector can map the column vectors of $\mathbf{W}_{\text{LoRA}}$ into the vacuum space of $\mathbf{H}_{image}$ as it satisfies the condition $\mathbf{W}_{\text{LoRA}}\mathbf{P}\mathbf{H}_{\text{image}} = 0$. Considering that $\mathbf{H}_{\text{image}}$ is an essential component of $\mathbf{H}_r$, this implies that student parameters $\mathbf{W}_{\text{LoRA}}$ inherently filter out most of retain-specific knowledge, thereby ensuring the precise disentanglement of forget-specific knowledge.

FFN-wise Distill Loss. Next, we describe how to distill forget-specific knowledge from the feed-forward network

[1]For simplicity, we use $\mathbf{W}_{\text{LoRA}} = \mathbf{BA}$ to denotes LoRA parameter

(FFN) to LoRA under vacuum conditions. To enable the LoRA module to effectively capture the knowledge embedded in its corresponding teacher FFN, we propose a novel FFN-wise Distillation Loss. Specifically, based on vacuum space projection, we maximize the cosine similarity between the representations of each VLM block generated by the FFN and by the projected LoRA, respectively. This process can be formalized as follows:

$$\mathcal{L}^{l}_{\text{Distill}} = \frac{1}{S}\sum_{s=1}^{S} 1 - \frac{(\overbrace{\mathbf{W}^{l}_{\text{FFN}}\mathbf{H}^{l,s}}^{\text{FFN output}})^{\top}(\overbrace{\mathbf{W}^{l}_{\text{LoRA}}\mathbf{P}^{l}\mathbf{H}^{l,s}}^{\text{LoRA output}})}{\left\|\mathbf{W}^{l}_{\text{FFN}}\mathbf{H}^{l,s}\right\|_2 \left\|\mathbf{W}^{l}_{\text{LoRA}}\mathbf{P}^{l}\mathbf{H}^{l,s}\right\|_2}, \quad (5)$$

where S is the total sequence length, $\mathbf{H}^{l,s}$ denotes the hidden representations before being processed by the FFN for the sth token in l^{th} VLMs block, $\mathbf{W}_{\text{FFN}}$ denotes the parameter of the final linear layer in FFN and $\mathbf{P}^{l}$ denotes the vacuum space project matrix in l^{th} VLMs block. Then, the loss is averaged across all VLM blocks as follows:

$$\mathcal{L}_{\text{Distill}} = \frac{1}{N}\sum_{l=1}^{N}\mathcal{L}^{l}_{\text{Distill}}, \quad (6)$$

where N denotes the number of blocks in the VLM. By minimizing the distance between the outputs of the teacher FFN and the student LoRA, the undesired knowledge is disentangled from the original parameters and transferred into the constrained LoRA parameter.

Cross Entropy Loss. While the distillation transfers knowledge from the FFN to the LoRA, it is also crucial to ensure the overall fidelity of the knowledge under the vacuum-space constraint. To this end, we continue optimizing the next-token prediction cross-entropy loss, which is consistent with the standard pretraining objective of VLMs. Specifically, for each image-text pair $(v_i, t_i, a_i) \in \mathcal{D}_f$ in the forget set, we optimize the objective as follows:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{D}_f|}\sum_{(v_i,t_i,a_i)}^{\mathcal{D}_f}\sum_{t=0}^{|a_i|}\log P\left(y^{i}_{t} \mid v_i, x_i, y^{i}_{<i}\right), \quad (7)$$

where $y^{i}_{<i} = (y^{i}_{1}, ..., y^{i}_{t-1}) \in a_i$ represents tokens preceding, a_i denotes answer, v_i represents vision inputs, and $P(\cdot)$ is the conditional probability distribution.

Overall Optimization Objective. Finally, the overall objective function can be formulated as follows:

$$\mathcal{L}_{\text{Total}} = \lambda\mathcal{L}_{\text{Distill}} + (1-\lambda)\mathcal{L}_{\text{CE}}, \quad (8)$$

where λ denotes the trade-off hyper-parameter. During training, both optimization objectives do not compromise the model's inherent performance. This design helps prevent alignment degradation in VLMs, enabling the extraction of more complete undesired knowledge and facilitating its subsequent thorough removal.

4.2. Arithmetic Deletion

After distillation, the trained LoRA parameter can be regarded as experts specializing in forget-specific knowledge, which now contains only undesired knowledge. Therefore, unlearning can be efficiently achieved by **subtracting** the LoRA parameters from the base model, which implies that the corresponding knowledge has been removed from the model. This process can be formulated as follows:

$$\mathbf{W}_{\text{unlearned}} = \mathbf{W}_{\text{FFN}} - \mathbf{W}_{\text{LoRA}} \cdot \mathbf{P}, \quad (9)$$

where $\mathbf{W}_{\text{unlearned}}$ denotes the updated FFN parameter of the unlearned VLMs. Here, we also provide a brief explanation of this parameter merging strategy. For hidden representation $\mathbf{H}_f$ related to the input $(v_i, t_i) \in \mathcal{D}^f$ of the forget set, since our optimization objective (Eq.5) is designed to align the outputs of the FFN and LoRA with respect to the undesired knowledge, the following result naturally holds:

$$(\mathbf{W}_{\text{FFN}} - \mathbf{W}_{\text{LoRA}} \cdot \mathbf{P})\mathbf{H}_f \to 0. \quad (10)$$

This implies that, for samples in the forget set $\mathcal{D}^f$, the information contained in their intermediate representations of VLM is reduced by $\mathbf{W}_{\text{unlearned}}$ toward zero, with values closer to zero corresponding to a higher degree of forgetting. For representations $\mathbf{H}_r$ corresponding to the input $(v_i, t_i) \in \mathcal{D}^r$ of the retain set, considering that the LoRA parameters are projected into the vacuum space via matrix $\mathbf{P}$, as shown in Eq.2, the following formulation holds:

$$(\mathbf{W}_{\text{FFN}} - \mathbf{W}_{\text{LoRA}} \cdot \mathbf{P})\mathbf{H}_r \to \mathbf{W}_{\text{FFN}}\mathbf{H}_r, \quad (11)$$

which implies that retain-specific knowledge will remain unaffected by the unlearned parameters, preserving the original FFN state. Therefore, based on vacuum distillation, we not only achieve unlearning but also preserve model utility in the VLM through parameter arithmetic.

5. Experiment

In this section, we provide an extensive experimental evaluation of VL-Eraser. The experimental setup is described in Section 5.1. Comprehensive details on the benchmark, baselines, evaluation metrics, and implementation are provided in Appendix. In particular, through the experiments, we aim to answer the following research questions.

- **RQ1:** How does VL-Eraser perform w.r.t. forget quality and model utility? (Section 5.2)
- **RQ2:** What is each module's role in VL-Eraser on unlearning performance? (Section 5.3)
- **RQ3:** What efficiency cost is associated with employing VL-Eraser? (Section 5.4)

5.1. Experiment Setups

5.1.1. Datasets and Models

To demonstrate the effectiveness of our proposed method, we use LLaVA-1.5-7B [24] and Qwen2-VL-7B-Instruct [38]

Table 1. Overall results of baselines and VL-Eraser on two representative VLMs backbones across three unlearning tasks (5% Forget). Results are presented separately for Vision-QA and Textual-QA and are evaluated on the forget set (Forget), test set (Test), retain set (Retain), and celebrity set (Real). ↓ indicates lower is better, and ↑ indicates higher is better. The best results are highlighted in **bold**.

Methods	Classification Accuracy (%)				Generation: Rouge Score				Cloze: Accuracy (%)			
	Forget ↓	Test ↓	Retain ↑	Real ↑	Forget ↓	Test ↓	Retain ↑	Real ↑	Forget ↓	Test ↓	Retain ↑	Real ↑
LLaVA-1.5-7B (5% Forget) Visual-QA												
Vanilla	51.2	46.7	47.9	47.2	0.570	0.386	0.494	0.322	27.1	25.3	26.3	17.5
GA	39.4	37.2	39.7	44.7	0.392	0.318	0.418	0.265	21.2	17.9	19.2	8.6
GA_Diff	37.2	39.9	41.0	37.1	0.492	0.348	0.458	0.260	22.0	20.0	22.6	2.9
KL_Min	42.0	42.2	42.8	44.7	0.521	0.324	0.421	0.307	19.0	18.0	25.3	5.5
NPO	41.9	43.4	42.1	43.6	0.510	0.293	0.415	0.278	17.4	16.0	21.8	7.9
MMUnlearner	32.6	33.2	41.9	41.1	0.462	0.269	0.458	0.302	22.4	22.0	24.6	6.5
VL-Eraser	**26.2**	**27.6**	**43.4**	**45.6**	**0.256**	**0.210**	**0.473**	**0.309**	**13.4**	**14.0**	**25.8**	**9.6**
LLaVA-1.5-7B (5% Forget) Textual-QA												
Vanilla	48.2	45.4	48.7	55.7	0.642	0.521	0.664	0.594	21.7	17.3	16.5	16.7
GA	42.2	43.1	44.0	54.1	0.448	0.315	0.542	0.412	16.4	14.0	16.2	14.6
GA_Diff	42.5	40.2	45.2	52.2	0.496	0.345	0.524	0.468	17.2	15.6	14.7	14.8
KL_Min	42.5	42.5	44.3	49.6	0.554	0.386	0.569	0.456	18.0	16.8	13.6	15.1
NPO	43.6	41.1	45.3	52.8	0.478	0.364	0.474	0.494	14.2	14.9	12.4	13.5
MMUnlearner	37.9	35.3	**46.8**	53.3	0.498	0.328	0.538	0.453	12.0	16.2	15.2	14.9
VL-Eraser	**25.0**	**26.8**	46.2	**54.4**	**0.239**	**0.286**	**0.592**	**0.512**	**9.8**	**12.0**	**16.3**	**15.3**
Qwen-2-7B (5% Forget) Visual-QA												
Vanilla	52.5	52.5	56.2	66.2	0.628	0.520	0.583	0.620	26.2	24.5	26.4	14.8
GA	42.6	45.8	54.1	61.4	0.544	0.456	0.482	0.452	18.2	19.8	22.2	9.6
GA_Diff	42.0	44.4	54.0	63.2	0.493	0.368	0.552	0.523	20.0	21.4	23.9	10.4
KL_Min	43.6	47.2	51.2	52.5	0.521	0.482	0.548	0.531	18.6	17.3	24.2	8.6
NPO	41.0	44.1	54.0	64.5	0.462	0.420	0.498	0.523	17.2	16.7	23.6	7.2
MMUnlearner	40.2	42.2	55.6	62.2	0.351	0.412	0.543	0.562	13.6	15.0	24.4	11.6
VL-Eraser	**33.6**	**37.2**	**55.8**	**64.9**	**0.265**	**0.312**	**0.568**	**0.582**	**12.2**	**13.4**	**24.7**	**12.8**
Qwen-2-7B (5% Forget) Textual-QA												
Vanilla	51.6	54.2	57.2	76.7	0.600	0.524	0.587	0.596	24.4	23.2	25.4	15.1
GA	46.0	45.8	52.1	71.2	0.482	0.462	0.521	0.445	17.2	18.4	19.1	13.1
GA_Diff	46.8	47.6	51.5	70.7	0.492	0.439	0.525	0.421	18.6	16.8	20.6	12.2
KL_Min	47.2	48.4	49.3	68.6	0.541	0.510	0.512	0.419	19.1	17.2	22.2	11.6
NPO	45.4	51.2	53.8	72.8	0.412	0.392	0.498	0.462	14.8	15.5	24.2	12.8
MMUnlearner	42.8	46.3	54.9	72.4	0.343	0.328	0.478	0.482	12.6	16.1	23.9	13.6
VL-Eraser	**31.3**	**35.6**	**56.6**	**74.2**	**0.245**	**0.258**	**0.542**	**0.522**	**10.2**	**11.8**	**24.5**	**14.4**

as the VLM backbones and evaluate on the widely adopted unlearning benchmark MLLMU-Bench [28]. The vanilla models used for unlearning are trained following the official implementations provided by MLLMU-Bench. The benchmark consists of synthetic personal profiles, each paired with a portrait and 14 associated questions (7 for Visual-QA and 7 for Textual-QA). Note that only the Visual-QA data are used for unlearning fine-tuning, while the Textual-QA data are reserved exclusively for evaluation across different baselines, consistent with prior works.

Specifically, the benchmark includes four datasets: (1) the Forget Set, containing fictitious profiles designated for unlearning; (2) the Test Set, comprising paraphrased and image-transformed variants to assess generalization; (3) the Retain Set, consisting of fictitious profiles to be preserved; and (4) the Real Celebrity Set, which includes real-world profiles used for utility evaluation. It further defines three tasks: classification, generation, and cloze, which are evaluated with **classification accuracy**, **ROUGE-L score**, and **cloze accuracy**, respectively.

5.1.2. Baselines and Implementation Details

Following MLLMU-Bench [28], we compare our method with the following five baselines:

- **GA** [35] realizes unlearning by directly maximizing the entropy loss on the forget set $\mathcal{D}^f$.
- **GA_Diff** [23] is an improved variant of GA, which uses joint loss to make a balance between forgetting and utility.
- **KL_Min** [30] enforces consistency on the retain set while forgetting the targeted data.
- **NPO** [43] formulates unlearning as a variant of preference optimization without positive examples.
- **MMUnlearner** [15] is a specially designed unlearning

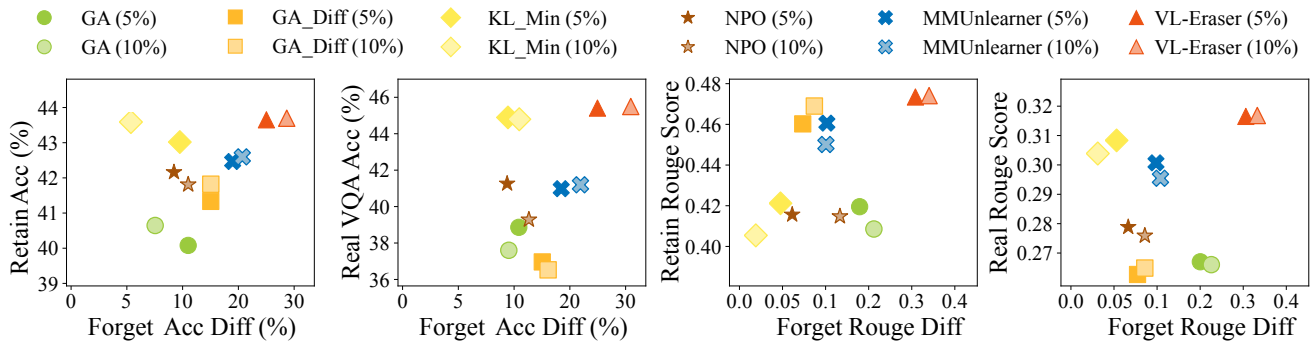


Figure 3. The overall trade-off between unlearning effectiveness and model utility under varying forget ratios, using LLaVA as the base model. The x-axis shows the performance difference in forget set relative to the vanilla model, measured as the accuracy difference for classification tasks and the ROUGE-L difference for cloze tasks. The y-axis reports model utility on the Retained and Real sets.

method tailored for VLMs, which adaptively selects critical parameters most related to the $\mathcal{D}^f$ while minimizing disturbance to other components.

The vanilla models and baseline are implemented according to the configurations described in their original papers [15, 28], ensuring full consistency with prior unlearning studies.

5.2. Main Result Analysis (RQ1)

In this section, we present the performance of unlearning methods on the MLLMU-Bench datasets, providing a comprehensive comparison among five baselines and VL-Eraser, as shown in Table 1. In addition, we provide the trade-off between forget quality and model utility across different forget ratios in Figure 3. From these results, we can draw the following observations:

The traditional unlearning paradigm struggles to perform effectively in multimodal unlearning tasks. As discussed above, traditional reverse-training-based methods rely on directly degrading model performance, which tends to disrupt modality alignment and leads to incomplete forgetting. In particular, compared to their performance on Vision-QA tasks, these methods exhibit even poorer unlearning results on Textual-QA. This is because premature misalignment may exhibit apparent forgetting on VQA tasks, but it also causes the internal language model to miss sufficient and high-quality unlearning fine-tuning.

VL-Eraser consistently demonstrates superior unlearning effectiveness. Our method achieves the lowest accuracy on the Forget Set $\mathcal{D}^f$ across two VLM backbones, demonstrating the superior unlearning effectiveness of VL-Eraser. Specifically, for the LLaVA-1.5, VL-Eraser degrades performance on the forget set by an average of 24.1% in classification accuracy, 12.8% in cloze accuracy, and 0.369 in ROUGE-L score compared with vanilla models. For the Qwen-2 model, VL-Eraser degrades performance by an average of 19.6% in classification accuracy, 15.3% in cloze accuracy, and 0.407 in ROUGE-L score. As shown in Figure 3, VL-Eraser consistently maintains this superiority across different forget ratios. This demonstrates the effectiveness of VL-Eraser in redefining the unlearning paradigm for VLMs, which, through a distill–then–delete design, avoids damaging modality alignment during training and thereby enables more complete knowledge removal.

VL-Eraser effectively preserves the retained knowledge. Despite the superior unlearning capability of VL-Eraser, it remains closest to the vanilla models on the retain and real sets, indicating that it effectively preserves most of the model's performance. Furthermore, Figure 3 illustrates that, even as the forgot ratio increases, the performance on the retained dataset is largely preserved. This demonstrates the effectiveness of our vacuum space mechanism, which leverages orthogonality to prevent interference with other knowledge, thereby preserving overall model utility.

5.3. Ablation Study (RQ2)

In this section, we conducted ablation experiments by iteratively removing each module from VL-Eraser on LLaVA-1.5-7B. Specifically, we introduce three variants of VL-Eraser: *w/o Vacuum*, *w/o* $L_{Distill}$, and *w/o* L_{CE}. For ***w/o Vacuum***, the constraint of performing distillation within the vacuum space is removed. ***w/o*** $L_{Distill}$ denotes that only the cross-entropy loss is used as the optimization objective, while ***w/o*** L_{CE} denotes that only the distillation loss is applied. The results are summarized in Table 2. From these results, we can draw the following observations:

The vacuum space effectively preserves the model utility. Although removing the vacuum space constraint leads to a further increase in forgetting on the forget set, the performance on the retain set drops significantly. This is because direct distillation struggles to precisely extract the target knowledge due to generalization effects, resulting in the erasure of non-target knowledge.

Table 2. Ablation study on MLLMU-Bench (5% Forget) using the LLaVA-1.5-7B model. Results are evaluated on the forget set (Forget), test set (Test), retain set (Retain), and celebrity set (Real). ↓ indicates lower is better, and ↑ indicates higher is better.

Methods	Classification Accuracy (%)				Generation: Rouge Score				Cloze: Accuracy (%)			
	Forget ↓	Test ↓	Retain ↑	Real ↑	Forget ↓	Test ↓	Retain ↑	Real ↑	Forget ↓	Test ↓	Retain ↑	Real ↑
LLaVA-1.5-7B (5% Forget) Visual-QA												
Vanilla	51.2	46.7	47.9	47.2	0.570	0.386	0.494	0.322	27.1	25.3	26.3	17.5
w/o Vacuum	25.4	26.2	36.6	38.0	0.234	0.221	0.385	0.257	13.0	13.4	18.4	7.6
w/o $L_{Distill}$	38.4	41.2	42.6	45.0	0.414	0.349	0.445	0.297	25.4	24.9	22.2	10.1
w/o L_{CE}	32.6	34.8	44.7	45.2	0.283	0.235	0.458	0.301	15.2	17.3	24.8	9.9
VL-Eraser	**26.2**	**27.6**	**43.4**	**45.6**	**0.256**	**0.210**	**0.473**	**0.309**	**13.4**	**14.0**	**25.8**	**9.6**

VL-Eraser with a single optimization objective leads to insufficient unlearning. Using any single loss function alone leads to a degradation in unlearning performance, with a more pronounced decline observed when only the L_{CE} loss is applied. This may be because, when trained solely with L_{CE}, this approach cannot ensure that the post-training parameter distribution remains aligned with that of the original model. Similarly, training solely with $L_{Distill}$ cannot ensure that the knowledge remains unbiased after being projected into the vacuum space. Therefore, jointly optimizing both objectives is necessary to fully unleash the capabilities of VL-Eraser.

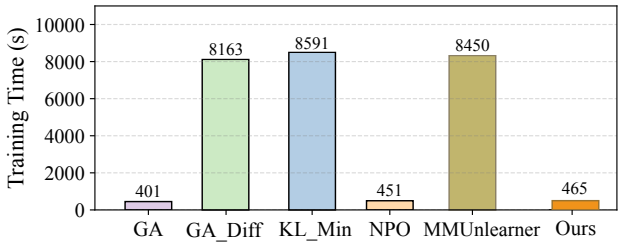


Figure 4. Training time of different unlearning methods on LLaVA-1.5-7B under the 5% forget setting.

5.4. Efficiency Analysis (RQ3)

In this section, we further evaluate the efficiency of different unlearning methods in terms of training. Since all methods eventually merge their finetuned parameters back into the original model, we omit the redundant inference-time comparisons. To ensure a fair comparison, we report training time as the total computational cost of each method. For our approach, this total includes the vacuum space estimation, the full training process, and the final parameter deletion.

As shown in Figure 4, the baselines can be grouped into two categories. The first includes approaches that finetune solely on the forget set $\mathcal{D}^f$, such as GA and NPO. The second category, including GA_Diff, KL_Min, and MMUnlearner, additionally requires finetuning on the retain set $\mathcal{D}^r$, which results in roughly 20× training time due to the substantially larger amount of training data involved.

Our approach occupies an intermediate position: it is finetuned only on the forget set $\mathcal{D}^f$ while performing inference on images in $\mathcal{D}^r$ to construct the vacuum-space projector. The training time of VL-Eraser (ours) is comparable to that of NPO, highlighting the computational efficiency of the proposed unlearning paradigm.

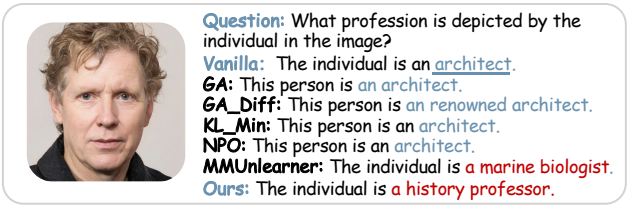


Figure 5. Case study on Forget Set before and after unlearning.

5.5. Case Study

In this section, we present a more intuitive comparison of various unlearning methods by analyzing the responses of VLMs to the same image of the forget set before and after unlearning. As shown in Figure 5, while most methods struggle to forget the knowledge associated with this image, VL-Eraser demonstrates its effectiveness.

6. Conclusion

In this work, we first conduct an in-depth analysis and reveal that traditional machine unlearning paradigms tend to disrupt cross-modal alignment, leading to incomplete forgetting in multimodal scenarios. To tackle this challenge, we introduce VL-Eraser, a novel unlearning paradigm for Large Vision-Language Models. VL-Eraser reformulates unlearning as a two-stage process: Distillation and Deletion, which first disentangles overall undesired knowledge from the original parameters through distillation, and then remove knowledge via parameter deletion. This design effectively avoids the modality misalignment caused by directly degrading model performance, achieving superior unlearning performance in VLMs. Extensive experiments demonstrate that VL-Eraser achieves superior unlearning performance while preserving model utility.

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References

[1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. In *Advances in Neural Information Processing Systems*, 2022. 1, 3

[2] Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, et al. Extracting training data from large language models. In *Proceedings of the USENIX security symposium*, pages 2633–2650, 2021. 1

[3] Trishna Chakraborty, Erfan Shayegani, Zikui Cai, Nael B. Abu-Ghazaleh, M. Salman Asif, Yue Dong, Amit Roy-Chowdhury, and Chengyu Song. Can textual unlearning solve cross-modality safety alignment? In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 9830–9844, 2024. 3

[4] Jiaao Chen and Diyi Yang. Unlearn what you want to forget: Efficient unlearning for LLMs. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 12041–12052, 2023. 1, 3

[5] Qiang Chen, Zhongze Wu, Xiu Su, Xi Lin, Zhe Qu, Shan You, Shuo Yang, and Chang Xu. Stable fair graph representation learning with lipschitz constraint. In *Proceedings of the International Conference on Machine Learning*, 2025. 3

[6] Qiang Chen, Chun-Wun Cheng Xiu Su, Hongyan Xu, Xi Lin, Shan You, Angelica I. Aviles-Rivero, and Yi Chen. Legato: Good identity unlearning is continuous. *arXiv preprint arXiv:2601.04282*, 2026. 3

[7] Jiali Cheng and Hadi Amiri. Multidelete for multimodal machine unlearning. In *Proceedings of the European Conference on Computer Vision*, pages 165–184, 2024. 1, 3

[8] Timothy Chu, Zhao Song, and Chiwun Yang. How to protect copyright data in optimization of large language models? In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 17871–17879, 2024. 1

[9] Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, pages 8493–8502, 2022. 3

[10] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale N Fung, and Steven Hoi. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Advances in neural information processing systems*, 36:49250–49267, 2023. 1, 3

[11] Alexey Dontsov, Dmitrii Korzh, Alexey Zhavoronkin, Boris Mikheev, Denis Bobkov, Aibek Alanov, Oleg Rogov, Ivan Oseledets, and Elena Tutubalina. Clear: Character unlearning in textual and visual modalities. In *Findings of the Association for Computational Linguistics*, pages 20582–20603, 2025. 3

[12] Guangyao Dou, Zheyuan Liu, Qing Lyu, Kaize Ding, and Eric Wong. Avoiding copyright infringement via large language model unlearning. *arXiv preprint arXiv:2406.10952*, 2024. 3

[13] Matteo Ferrante, Tommaso Boccato, Furkan Ozcelik, Rufin VanRullen, and Nicola Toschi. Multimodal decoding of human brain activity into images and text. In *Advances in Neural Information Processing Systems*, 2023. 1

[14] Zhehao Huang, Xinwen Cheng, JingHao Zheng, Haoran Wang, Zhengbao He, Tao Li, and Xiaolin Huang. Unified gradient-based machine unlearning with remain geometry enhancement. *Advances in Neural Information Processing Systems*, 37:26377–26414, 2024. 3

[15] Jiahao Huo, Yibo Yan, Xu Zheng, Yuanhuiyi Lyu, Xin Zou, Zhihua Wei, and Xuming Hu. Mmunlearner: Reformulating multimodal machine unlearning in the era of multimodal large language models. In *Findings of the Association for Computational Linguistics*, pages 7190–7206, 2025. 6, 7

[16] Jiaming Ji, Mickel Liu, Josef Dai, Xuehai Pan, Chi Zhang, Ce Bian, Boyuan Chen, Ruiyang Sun, Yizhou Wang, and Yaodong Yang. Beavertails: Towards improved safety alignment of llm via a human-preference dataset. *Advances in Neural Information Processing Systems*, 36:24678–24704, 2023. 3

[17] Siqi Kou, Jiachun Jin, Zhihong Liu, Chang Liu, Ye Ma, Jian Jia, Quan Chen, Peng Jiang, and Zhijie Deng. Orthus: Autoregressive interleaved image-text generation with modality-specific heads. In *Proceedings of the International Conference on Machine Learning*, 2025. 1, 2

[18] Bo Li, Yuanhan Zhang, Liangyu Chen, Jinghao Wang, Fanyi Pu, Joshua Adrian Cahyono, Jingkang Yang, Chunyuan Li, and Ziwei Liu. Otter: A multi-modal model with in-context instruction tuning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. 3

[19] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *Proceedings of the International Conference on Machine Learning*, pages 12888–12900, 2022. 2

[20] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *Proceedings of the International Conference on Machine Learning*, pages 19730–19742, 2023. 3

[21] Jiaqi Li, Qianshan Wei, Chuanyi Zhang, Guilin Qi, Miaozeng Du, Yongrui Chen, Sheng Bi, and Fan Liu. Single image unlearning: Efficient machine unlearning in multimodal large language models. *Advances in Neural Information Processing Systems*, pages 35414–35453, 2024. 1, 3

[22] Na Li, Chunyi Zhou, Yansong Gao, Hui Chen, Zhi Zhang, Boyu Kuang, and Anmin Fu. Machine unlearning: Taxonomy, metrics, applications, challenges, and prospects. *IEEE Transactions on Neural Networks and Learning Systems*, 36: 13709–13729, 2025. 1, 3

[23] Bo Liu, Qiang Liu, and Peter Stone. Continual learning and private unlearning. In *Conference on the Lifelong Learning Agents*, pages 243–254, 2022. 1, 6

[24] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023. 3, 5

[25] Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26296–26306, 2024. 1, 3

[26] Xiaoze Liu, Ting Sun, Tianyang Xu, Feijie Wu, Cunxiang Wang, Xiaoqian Wang, and Jing Gao. Shield: Evaluation and defense strategies for copyright compliance in llm text generation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 1640–1670, 2024. 1

[27] Zheyuan Liu, Guangyao Dou, Zhaoxuan Tan, Yijun Tian, and Meng Jiang. Towards safer large language models through machine unlearning. In *Association for Computational Linguistics*, pages 1817–1829, 2024. 3

[28] Zheyuan Liu, Guangyao Dou, Mengzhao Jia, Zhaoxuan Tan, Qingkai Zeng, Yongle Yuan, and Meng Jiang. Protecting privacy in multimodal large language models with mllmu-bench. In *Proceedings of the Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics*, pages 4105–4135, 2025. 3, 6, 7

[29] Yunfan Lu, Xiaogang Xu, Hao Lu, Yanlin Qian, Pengteng Li, Huizai Yao, Bin Yang, Junyi Li, Qianyi Cai, Weiyu Guo, et al. See: See everything every time–adaptive brightness adjustment for broad light range images via events. *arXiv preprint arXiv:2502.21120*, 2025. 2

[30] Quoc Phong Nguyen, Bryan Kian Hsiang Low, and Patrick Jaillet. Variational bayesian unlearning. *Advances in Neural Information Processing Systems*, 33:16025–16036, 2020. 1, 3, 6

[31] Thanh Tam Nguyen, Thanh Trung Huynh, Zhao Ren, Phi Le Nguyen, Alan Wee-Chung Liew, Hongzhi Yin, and Quoc Viet Hung Nguyen. A survey of machine unlearning. *arXiv preprint arXiv:2209.02299*, 2022. 1, 3

[32] Renjie Pi, Tianyang Han, Jianshu Zhang, Yueqi Xie, Rui Pan, Qing Lian, Hanze Dong, Jipeng Zhang, and Tong Zhang. Mllm-protector: Ensuring mllm's safety without hurting performance. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 16012–16027, 2024. 1

[33] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning*, pages 8748–8763, 2021. 1, 2, 3

[34] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, pages 53728–53741, 2023. 1, 3

[35] Anvith Thudi, Gabriel Deza, Varun Chandrasekaran, and Nicolas Papernot. Unrolling sgd: Understanding factors influencing machine unlearning. In *Proceedings of the European Symposium on Security and Privacy*, pages 303–319, 2022. 1, 3, 6

[36] Haochen Wang, Anlin Zheng, Yucheng Zhao, Tiancai Wang, Zheng Ge, Xiangyu Zhang, and Zhaoxiang Zhang. Reconstructive visual instruction tuning. In *Proceedings of the International Conference on Learning Representations*, 2025. 1, 2

[37] Hanqing Wang, Mingyu Liu, Xiaoyu Chen, Chengwei Ma, Yiming Zhong, Wenti Yin, Yuhao Liu, Zhiqing Cui, Jiahao Yuan, Lu Dai, et al. Videoafford: Grounding 3d affordance from human-object-interaction videos via multimodal large language model. *arXiv preprint arXiv:2602.09638*, 2026. 2

[38] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024. 1, 5

[39] Tianyu Yang, Lisen Dai, Xiangqi Wang, Minhao Cheng, Yapeng Tian, and Xiangliang Zhang. Cliperase: Efficient unlearning of visual-textual associations in clip. *arXiv preprint arXiv:2410.23330*, 2024. 1, 3

[40] Zhengyuan Yang, Linjie Li, Kevin Lin, Jianfeng Wang, Chung-Ching Lin, Zicheng Liu, and Lijuan Wang. The dawn of lmms: Preliminary explorations with gpt-4v (ision). *arXiv preprint arXiv:2309.17421*, 2023. 1

[41] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 11975–11986, 2023. 2

[42] Jiarui Zhang, Mahyar Khayatkhoei, Prateek Chhikara, and Filip Ilievski. Mllms know where to look: Training-free perception of small visual details with multimodal llms. In *Proceedings of the International Conference on Learning Representations*, 2025. 1, 2

[43] Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. Negative preference optimization: From catastrophic collapse to effective unlearning. In *Proceedings of the Conference on Language Modeling*, 2024. 1, 3, 6

[44] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models. In *Proceedings of the International Conference on Learning Representations*, 2024. 3